Adaptive Meal Preparation in a Campus Canteen

A tabular Q-learning formulation for uncertain student demand

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GITHUB: https://github.com/Shahid20-K/AI-A2-Canteen-qlearning.git

1. Problem statement

A campus canteen must prepare enough meals to meet changing student demand without preparing so much that perishable food is discarded. Fixed preparation plans may respond poorly to day-to-day demand changes. This project studies whether a learned policy can balance availability and waste in a controlled simulation.

2. Environment and demand

One day has eight time slots with nominal mean demand **(8, 12, 22, 30, 24, 18, 27, 14)**. The hidden low/normal/high regime has multipliers 0.70/1.00/1.30 and prior probabilities 0.25/0.50/0.25. The agent sees a noisy category forecast: it is correct with probability 70%; when wrong, one of the other categories is selected uniformly. Demand is Poisson-distributed around the hidden regime-adjusted mean.

The inventory capacity is 40 portions. Of unsold portions, 30% (rounded down) can carry into the next slot; the remainder is discarded. Nothing carries beyond the final slot. These are explicit synthetic assumptions, not measured canteen data.

At each step, available stock equals starting inventory plus actual preparation, capped at capacity. Sales are the minimum of available stock and demand; unmet demand is the difference when demand exceeds stock. Carryover and discarded waste are then calculated before the next slot.

3. State space S

The agent observes **(time slot, exact inventory, noisy daily outlook, sales pace, prior-stockout flag)**. Sales pace is below/on/above expected cumulative sales from completed slots; the stockout flag warns that low sales may reflect unavailable food. The true regime remains hidden, and this compact history summary is not a full belief state.

State count = 8 × 41 × 3 × 3 × 2 = 5,904 discrete states.

The sales-pace category compares cumulative sales with cumulative posterior-expected demand: below 85% is below expected, above 115% is above expected, otherwise sales are on track. A prior-stockout flag helps identify censored sales. Demand remains uncertain.

4. Action space A

At each slot, the agent selects an additional preparation batch from **{0, 5, 10, 15, 20}** portions. The 20-portion per-slot maximum is an assumed batch-size limit, not a measured kitchen capacity. Separately, total inventory is capped at 40 portions.

5. Episode objective and hypothesis

One episode is a complete simulated day of eight decisions; its return is the sum of slot rewards. The experiment tests whether a state-dependent learned policy can outperform static preparation rules when the daily demand outlook changes. The policy is not assumed to succeed: this claim is assessed using held-out simulated episodes and operational metrics.

6. Slot transition

For slot t, available stock is $A_t = \min(I_t + p_t, 40)$. Sales are $\min(A_t, D_t)$ and unmet demand is $\max(D_t - A_t, 0)$. Leftover stock is $\max(A_t - D_t, 0)$; carryover is floor(0.30 × leftover) except at the final slot, where it is zero. Discarded waste equals leftover minus carryover.

The hidden regime persists for the day while the forecast is noisy. Sales pace and prior stockouts summarize earlier outcomes but do not form a full Bayesian belief state, so the process remains partially observable.

7. Reward function R

The reward encourages serving demand while accounting for preparation, stockouts, and food discarded:

$$r = \text{sold} - 0.35 \times \text{prepared} - 1.00 \times \text{unmet demand} - 1.50 \times \text{discarded waste}$$

- **sold:** portions served to students.
- **prepared:** additional portions actually produced in the slot.
- **unmet demand:** demand not met because inventory was insufficient.
- **discarded waste:** unsold portions that do not carry over.

The coefficients are illustrative utility weights, not monetary estimates. They make competing objectives explicit. Changing these weights can change the policy, so they should be reported and justified rather than tuned invisibly.

8. Learning algorithm

We use tabular Q-learning with epsilon-greedy exploration. The update is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Terminal transitions do not bootstrap future value. Each final agent trains for 6,000 episodes with $\alpha = 0.08$, $\gamma = 0.92$, and ε decay = 0.9992; ε starts at 1.0 and floors at 0.05. A four-candidate, two-seed validation sweep selected α and decay before held-out testing. Five final training seeds assess robustness.

Early in training, epsilon-greedy exploration tries different preparation batches; as epsilon decays, the agent increasingly uses actions with the highest current Q-value. Discounting values rewards that occur later in the simulated day, so a decision can account for carryover into the next slot.

9. Evaluation design

Each policy is tested on the same 2,000 held-out simulated days using matched seeds. The rule baseline uses the same forecast, sales-pace category, and prior-stockout flag; its target (selected from 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%) and history correction are tuned on 400 validation days. Q-learning's learning rate and ε decay are tuned separately on a two-seed validation sweep. Five final training seeds, paired intervals, and 0.75×/1.25× stress tests assess robustness.

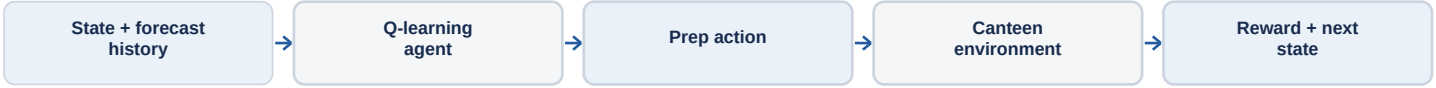
10. Reproducibility

The primary agent uses seed 2026; four more agents use independent seeds. Test episodes are separate from training and use matched demand draws across policies. Outputs include the Q-table, returns, metrics JSON, plots, policy slices, and an example trace. Twenty unit tests cover forecast inference, sales-history state encoding, transitions, rewards, configuration, training, and reproducibility.

11. Scope of evidence

The baseline comparison supports conclusions only about this simulation and reward design. It does not establish that the same improvement would occur in a live canteen. Real deployment would require calibrated demand data, cost estimates, and operational and food-safety constraints.

RL decision cycle



The next state and reward feed into the Q-value update; the loop repeats until the simulated day ends.

Experimental Results and Discussion

Synthetic simulation results, interpretation, and limitations

Synthetic simulation only; no real canteen records used.

Main test: 2,000 held-out synthetic days per policy. Across five independently trained Q-learning agents, mean reward was 24.91 +/- 0.39 SD; the table shows seed 2026 for paired test-day comparisons. These are simulated outcomes, not real-canteen measurements.

Policy	Mean reward/day	Sold/day	Unmet demand/day	Waste/day
Q-learning (seed 2026)	25.30	118.04	37.72	7.41
Fixed 15-portion batch	-5.08	106.48	49.29	13.52
Forecast-aware history rule	26.60	121.93	33.84	10.17

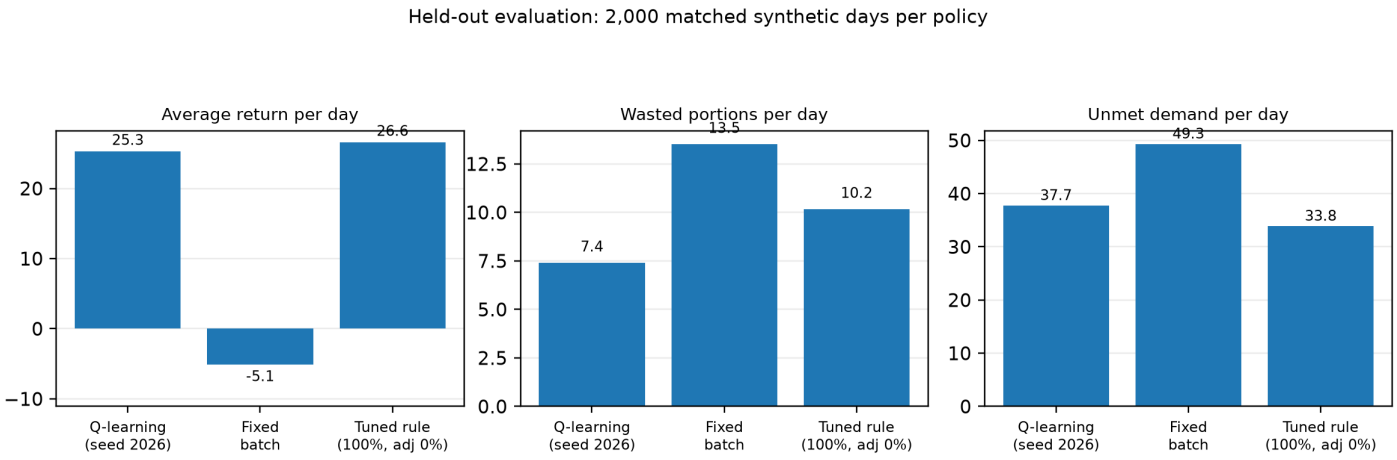


Figure 1. Mean reward, waste, and unmet demand for the learned policy and comparison baselines.

12. Learning behavior

The curve plots 200-episode return means averaged across five training seeds, with a ±1 SD band. Returns rise early and then fluctuate around a stable range. This is evidence of training progress in the simulator, not proof of real-world performance.

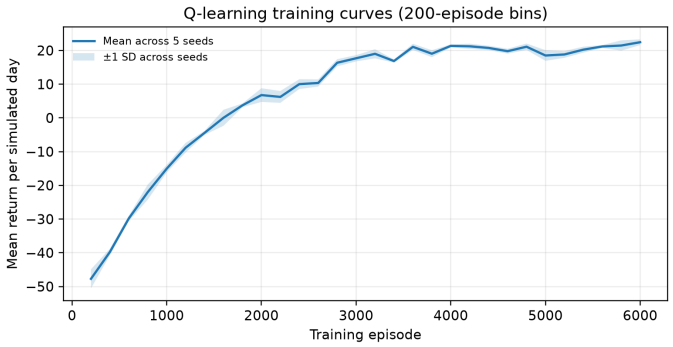


Figure 2. Mean return in 200-episode bins across five seeds; shaded band is ±1 SD.

13. Interpretation and paired checks

On the held-out test set, Q-learning reward was 25.30 (95% CI 24.56 to 26.05) versus 26.60 for the tuned schedule (95% CI 25.81 to 27.38). The paired reward difference (Q-learning minus baseline) was -1.29 (95% CI -1.66 to -0.93); versus fixed-batch it was 30.38 (29.77 to 30.99). Q-learning discarded fewer portions (7.41 vs 10.17) but sold fewer and had more unmet demand than the tuned schedule. Across five training seeds, mean Q-learning reward was 24.91 (SD 0.39).

Under demand shifts, mean reward for Q-learning versus the nominal-planning tuned schedule was 8.60 vs 12.69 at 0.75× demand, and 3.53 vs 9.03 at 1.25×. Both policies keep their nominal planning assumptions; the results show sensitivity to a changed environment.

14. Limitations

Demand is synthetic and Poisson-distributed. The environment assumes one meal type, eight time slots, a forecast with assumed 70% accuracy, a 20-portion maximum action, simplified carryover, and illustrative reward weights. Sales pace and prior stockouts summarize history but do not maintain a full belief over the persistent hidden regime. No real-world savings are established.

15. Reward-weight sensitivity

Train waste wt.	Base reward/day	Waste/day	Unmet/day
0.75	24.77 +/- 0.50	8.23 +/- 0.37	37.23 +/- 0.62
1.50	23.93 +/- 1.51	6.55 +/- 1.02	39.63 +/- 2.05
2.25	20.99 +/- 0.85	4.61 +/- 0.10	43.57 +/- 0.52

Each setting trains three seeds; each policy is tested on the same 2,000 held-out scenarios using the original reward weights. Values are mean +/- SD across trained seeds, not confidence intervals. As the training waste penalty rises, waste falls but unmet demand rises. This demonstrates reward sensitivity, not an objectively optimal coefficient.

16. Future work

Future work could maintain a richer belief state, test different forecast accuracies, calibrate assumptions with permitted anonymized records, and model freshness, lead times, and food-safety constraints.

Conclusion

Within this noisy-forecast simulation, Q-learning is compared with a fixed batch and a history-aware forecast rule using the same observations. The sales-pace and prior-stockout features provide a compact summary of past outcomes but not a full belief state. Results are simulator-specific and show trade-offs rather than a universal RL win.

Reproducibility: source, trained Q-table, tests, metrics, sensitivity analysis, and figure scripts are included in this repository.